Taller 3

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1)

$$\dot{x} = \begin{pmatrix} -x_1^3 + x_2 \\ x_1 x_3 \\ -x_1 + x_3 \\ x_2 \\ x_5 + x_3^2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 2 \\ 0 \\ x_2 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} u_2, \quad \text{and} \quad y = \begin{pmatrix} x_3^2 \\ x_5 \end{pmatrix}$$

$$\dot{x}_1 = -x_1^3 + x_2$$
$$\dot{x}_2 = x_1 x_3 + u_1 + u_2$$
$$\dot{x}_3 = -x_1 + x_3 + 2u_1$$
$$\dot{x}_4 = x_2 + 2u_2$$
$$\dot{x}_5 = x_5 + x_3 + x_2 u$$

$$0 = -x_1^3 + x_2$$
$$0 = x_1 x_3 + u_1 + u_2$$
$$0 = -x_1 + x_3 + 2u_1$$
$$0 = x_2 + 2u_2$$
$$0 = x_5 + x_3 + x_2 u$$

$$\begin{bmatrix} \dot{x}_1 \delta \\ \dot{x}_2 \delta \\ \dot{x}_3 \delta \\ \dot{x}_4 \delta \\ \dot{x}_5 \delta \end{bmatrix} = \underbrace{\begin{bmatrix} -3x_1^2 & 1 & 0 & 0 & 0 \\ x_3 & 0 & x_1 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & u_1 & 2x_3 & 0 & 1 \end{bmatrix}}_{A} + \underbrace{\begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 2 & 0 \\ 0 & 2 \\ x_1 & 0 \end{bmatrix}}_{B} \quad u_1, u_2$$

$$C = \begin{bmatrix} 0 & 0 & 2x_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$x_{eq} = \begin{vmatrix} 0 & 0 & 0 & 0 & 0 \end{vmatrix}$$

$x_1 = 2 \qquad x_3 = 2$

$$\underbrace{\begin{vmatrix} -12 & 1 & 0 & 0 & 0 \\ 2 & 0 & 2 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 1 \end{vmatrix}}_{A} \qquad \underbrace{\begin{vmatrix} 0 & 0 \\ 1 & 1 \\ 2 & 0 \\ 0 & 2 \\ 8 & 0 \end{vmatrix}}_{B}$$

$$\underbrace{\begin{vmatrix} 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{vmatrix}}_{C} \qquad \underbrace{\begin{vmatrix} 0 & 0 \end{vmatrix}}_{D}$$

Desacople

$V_1 = 1$

$$\begin{vmatrix} 0 & 0 & 4 & 0 & 0 \end{vmatrix} \begin{vmatrix} 0 & 0 \\ 1 & 1 \\ 2 & 0 \\ 0 & 2 \\ 8 & 0 \end{vmatrix} = \begin{vmatrix} 8 & 0 \end{vmatrix}$$

$V_2 = 1$

$$\begin{vmatrix} 0 & 0 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} 0 & 0 \\ 1 & 1 \\ 2 & 0 \\ 0 & 2 \\ 8 & 0 \end{vmatrix} = \begin{vmatrix} 8 & 0 \end{vmatrix}$$

$$H_{LA} = \begin{vmatrix} 1/s^2 & 0 \\ 0 & 1/s^2 \end{vmatrix} \qquad \epsilon = \begin{vmatrix} 8 & 0 \\ 8 & 0 \end{vmatrix} \qquad P = \begin{vmatrix} -4 & 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 & 1 \end{vmatrix}$$

$$F = \epsilon^{-1} P \qquad F = \begin{vmatrix} -32 & 0 & 32 & 0 & 0 \\ -32 & 0 & 32 & 0 & 0 \end{vmatrix}$$

$F(s) = \frac{1}{s^2}$ $\quad e_{ss} = 0$ $\quad \sqcap \quad t_{sd} = 1 \quad \xi_d = 0{,}7 \quad w_{nd} = \frac{4}{1 \cdot 0{,}7} = 5{,}71$

$T_m = \frac{1}{20} = 0{,}05$

$$\frac{0{,}00125\, z + 0{,}00125}{z^2 - 2z + 1} \Rightarrow \frac{0{,}00125\, z^{-1} + 0{,}00125\, z^{-2}}{1 - 2z^{-1} + z^{-2}}$$

FCO:

$$G = \begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} \qquad H = \begin{vmatrix} 0{,}00125 \\ 0{,}00125 \end{vmatrix}$$

$$C = \begin{vmatrix} 0 & 1 \end{vmatrix} \qquad D = 0$$

$$\hat{G} = \begin{vmatrix} 0 & -1 & 0 \\ 1 & 2 & 0 \\ -1 & -2 & 1 \end{vmatrix} \qquad \hat{H} = \begin{vmatrix} 0{,}0013 \\ 0{,}0012 \\ -0{,}0013 \end{vmatrix}$$

$$\hat{S} = [\hat{H} \;\; \hat{G}\hat{H} \;\; \hat{G}^2\hat{H}] = \begin{vmatrix} 0{,}0013 & -0{,}0013 & -0{,}0037 \\ 0{,}0013 & 0{,}0037 & 0{,}0063 \\ -0{,}0016 & -0{,}005 & -0{,}0112 \end{vmatrix}$$

$s^2 + 8s + 32{,}653$

$s = -4 \pm 4{,}08j$

$$P_1 = 2e^{-\xi w_n T_m} \cos(w_n T_m \sqrt{1-\xi^2}) = -1{,}6035$$

$$P_2 = e^{-2\xi w_n T_m} = 0{,}6703$$

$(z^2 + P_1 z + P_2)(z - 0{,}1)$

$z^3 - 1{,}7 z^2 + 0{,}831 z - 0{,}067$

$$\phi G = \begin{vmatrix} -0{,}367 & -0{,}431 & 0 \\ 0{,}431 & 0{,}495 & 0 \\ -1{,}731 & -2{,}162 & 0 \end{vmatrix} \qquad K = \begin{vmatrix} 440{,}2 & 574{,}2 & -25{,}6 \end{vmatrix}$$

Observador

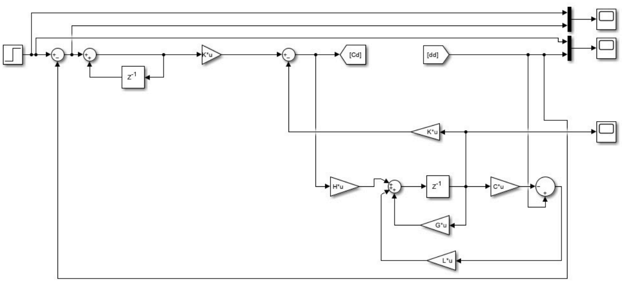
$t_{so} = \frac{1}{10} = 0,1$ $\qquad$ $\omega_{ndo} = \frac{4}{0,1 \cdot 0,7} = 57,1429$

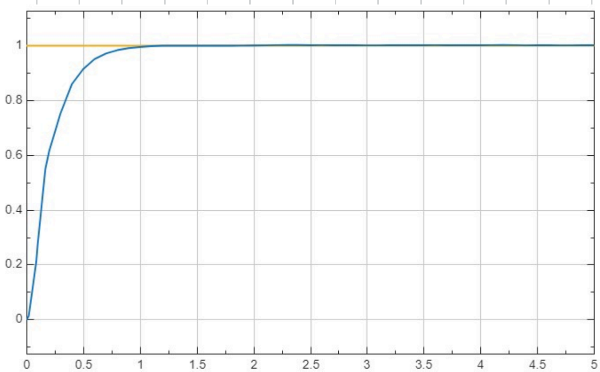
$s^2 + 80s + 3265,3$

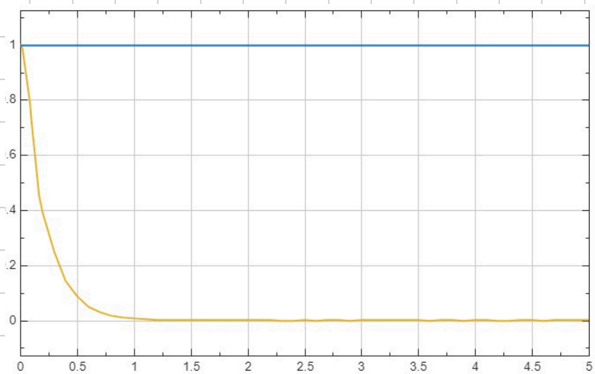
$P_{1o} = 0,1225$
$P_{2o} = 0,0183$

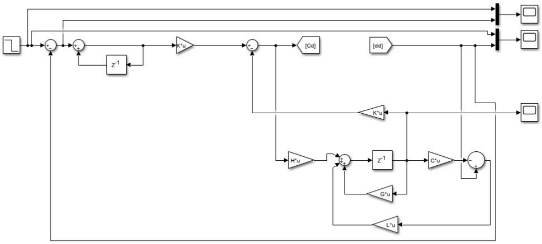
$z^2 + 0,122\,z + 0,0183$

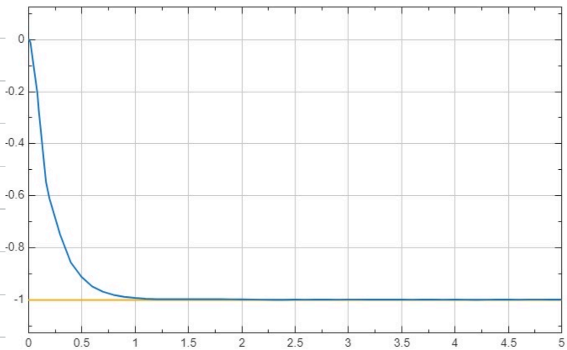
$$L = \begin{vmatrix} -0,9817 \\ 2,1225 \end{vmatrix}$$








z-1
K*u
[Cd]
[dd]
K*u
H*u
z-1
C*u
G*u
L*u


0
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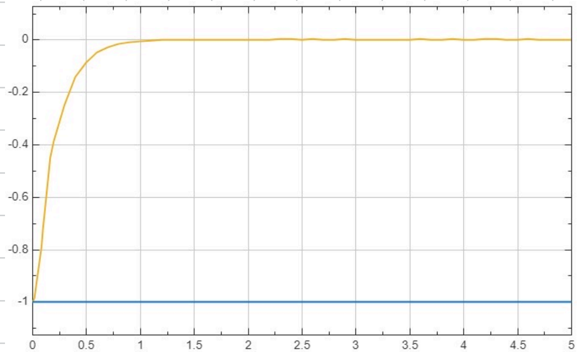
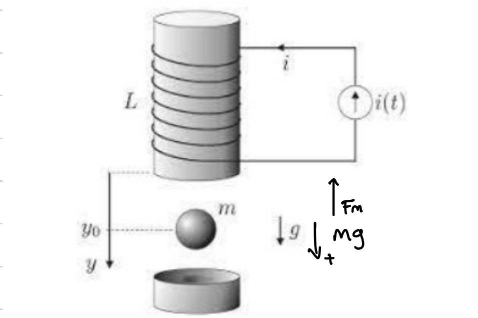
-0.4
-0.6
-0.8
-1
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5


0
-0.2
-0.4
-0.6
-0.8
-1
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5

2)



$$F_{m(i,y)} = k\frac{i^2}{y^2}$$

$$m\ddot{y} = mg - F_m = m\ddot{y} = mg - k\frac{i^2}{y^2}$$

$$\ddot{y} = g - \frac{k}{m}\frac{i^2}{y^2} \qquad v = \dot{y} \qquad \dot{v} = g - \frac{K}{m}\frac{i^2}{y^2}$$

Puntos de equilibrio $(\ddot{y} = \dot{y} = 0)$

$$0 = g - \frac{K}{m}\cdot\frac{i^2}{y^2}$$

$$g = \frac{K}{m}\cdot\frac{i^2}{y^2} \longrightarrow \frac{K}{m} = \boxed{\frac{gy^2}{i^2}}$$ Punto operación

Taylor

$$f(y,i) = g - \frac{k}{m}\cdot\frac{i^2}{y^2} \qquad \delta\ddot{y} = \frac{\frac{\partial f}{\partial y}}{\delta y} + \frac{\frac{\partial f}{\partial i}}{\partial i}$$

$$\frac{\partial f}{\partial y} = -\frac{k}{m}i^2\cdot(-2y^{-3}) = \frac{2ki^2}{my^3} = \frac{2i^2}{y^3}\cdot\frac{g\cdot y^2}{i^2} = \frac{2g}{y_0}$$

$$\frac{\partial f}{\partial i} = -\frac{k}{m}\cdot\frac{1}{y^2}\cdot 2i = \frac{-2ki}{my^2} = \frac{-2i}{y^2}\cdot\frac{g\cdot y^2}{i^2} = \frac{-2g}{i^2}$$

$$\delta\ddot{y} = \frac{2g}{y_0}\delta_y - \frac{2g}{i^2}\delta i$$

Matriz de estados

$$\begin{bmatrix} \delta\dot{y} \\ \delta\ddot{y} \end{bmatrix} = \underbrace{\begin{vmatrix} 0 & 1 \\ \frac{2g}{y_0} & 0 \end{vmatrix}}_{A} \begin{vmatrix} \delta y \\ \delta\dot{y} \end{vmatrix} + \underbrace{\begin{vmatrix} 0 \\ \frac{-2g}{i} \end{vmatrix}}_{B} \delta i$$

$$y = \underbrace{\begin{vmatrix} 1 & 0 \end{vmatrix}}_{C} \begin{vmatrix} \delta y \\ \delta\dot{y} \end{vmatrix} + \underbrace{\begin{vmatrix} 0 \end{vmatrix}}_{D} \delta i$$

$g = 9{,}81\ m/s^2$; $i = 19{,}62A$; $y_0 = 19{,}62m$

$$G(s) = \frac{y(s)}{I(s)} = \frac{-2g/i}{s^2 - 2g/y_0}$$

$$G(s) = \frac{-1}{s^2 - 1}$$

$e_{ss} = 0$ $\quad \zeta = 0{,}5$ $\quad t_{sd} = 5$ $\quad w_{nd} = 1{,}6$

$$t_m = \frac{5}{20} = 0{,}25$$

$$\frac{-0{,}03141z - 0{,}03141}{z^2 - 2{,}063z + 1} \Rightarrow \frac{-0{,}03141z^{-1} - 0{,}03141z^{-2}}{1 - 2{,}063z^{-1} + z^{-2}}$$

$$G = \begin{vmatrix} 0 & -1 \\ 1 & 2{,}063 \end{vmatrix} \quad H = \begin{vmatrix} -0{,}03141 \\ -0{,}03141 \end{vmatrix} \quad C = \begin{vmatrix} 0 & 1 \end{vmatrix} \quad D = 0$$

$$\hat{G} = \begin{vmatrix} 0 & -1 & 0 \\ 1 & 2{,}063 & 0 \\ -1 & -2{,}063 & 1 \end{vmatrix} \quad \hat{H} = \begin{vmatrix} -0{,}0314 \\ -0{,}0314 \\ 0{,}0314 \end{vmatrix} \quad \hat{S} = \begin{vmatrix} \hat{H} & \hat{G}\hat{H} & \hat{G}^2\hat{H} \end{vmatrix}$$

$$\hat{S} = \begin{vmatrix} -0{,}0314 & 0{,}0314 & 0{,}0962 \\ -0{,}0314 & -0{,}0962 & -0{,}1647 \\ 0{,}0314 & 0{,}1276 & 0{,}2947 \end{vmatrix}$$

$$s^2 + 1{,}6s + 2{,}56$$
$$s = -0{,}8 \pm 1{,}385j$$

$P_1 = 2e^{-\zeta w_n T_m} \cos(w_n T_m \sqrt{1-\zeta^2}) = -1{,}5402$

$P_2 = e^{-2\zeta w_n T_m} = 0{,}6703$

$(z^2 + P_1 z + P_2)(z - 0{,}1)$

$z^3 - 1{,}64 z^2 + 0{,}824 z - 0{,}067$

$$\phi G = \begin{vmatrix} -0{,}490 & -0{,}696 & 0 \\ 0{,}696 & 0{,}947 & 0 \\ -2{,}119 & -2{,}949 & 0 \end{vmatrix} \qquad K = \begin{vmatrix} -18{,}0028 & -25{,}4387 & 1{,}8625 \end{vmatrix}$$

Observador

$t_{so} = \dfrac{5}{10} = 0{,}5$ $\qquad w_{do} = \dfrac{4}{\zeta d \cdot t_{so}} = 16$

$s^2 + 16s + 256$

$s = -8 \pm 13{,}856j$

$P_1 D_o = 0{,}2567$

$P_2 D_o = 0{,}0183$

$z^2 + 0{,}257 z + 0{,}0183$

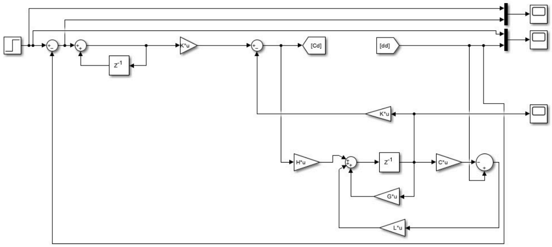
$$L = \begin{vmatrix} -0{,}9817 \\ 2{,}3197 \end{vmatrix}$$

$s^2 + 8s + 32{,}653$

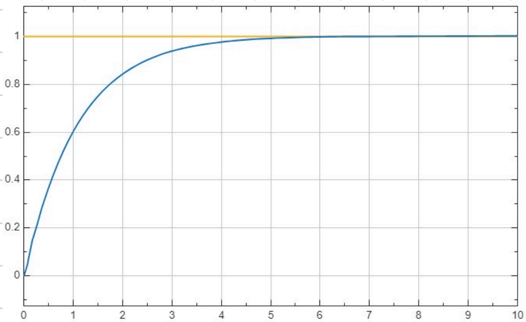
$s = -4 \pm 4{,}08j$

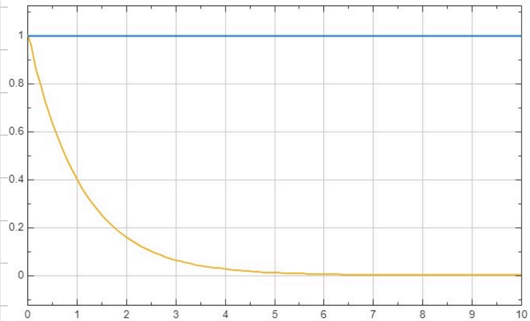
$P_1 = 2e^{-\zeta w_n T_m} \cos(w_n T_m \sqrt{1-\zeta^2}) = -1{,}6035$

$P_2 = e^{-2\zeta w_n T_m} = 0{,}6703$



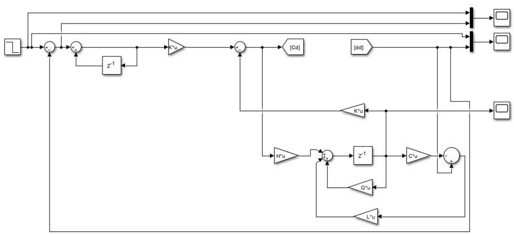
= Desacople y Observador

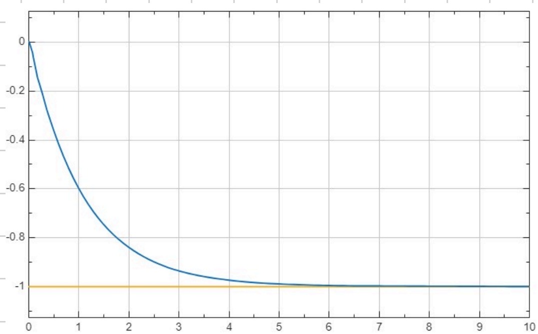


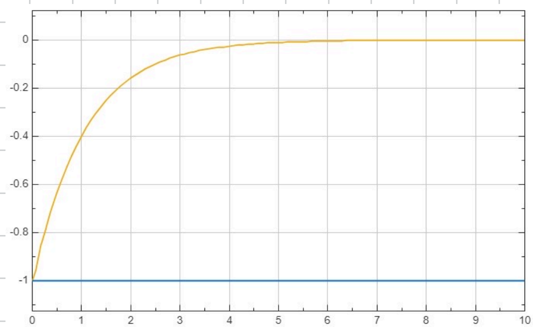


Step = 1 tsd = 5 s

error

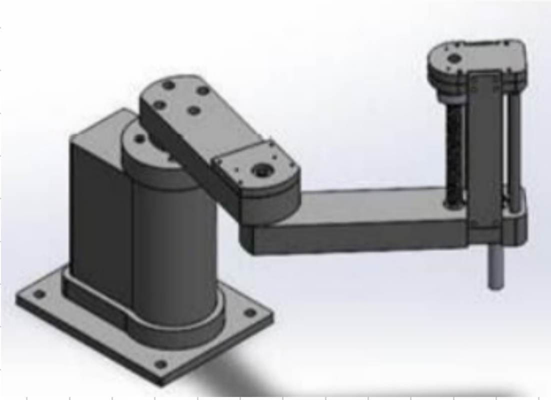


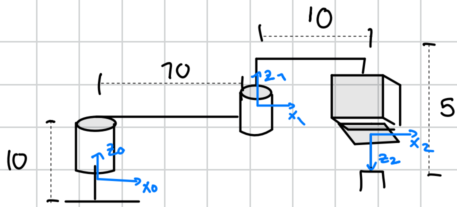




Step = -1 tsd = 5 s

3)





DH

n	θ	d	α	a
1	q_1+0	10	10	0
2	q_2+0	0	5	π
3	0	q_3	0	0

$q_1 = \theta_1$
$q_2 = \theta_2$
$q_3 = d_3$

$$T_0^3 = \begin{vmatrix} \cos(\theta_1+\theta_2) & \sin(\theta_1+\theta_2) & 0 & 10\cos(\theta_1)+5\cos(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & -\cos(\theta_1+\theta_2) & 0 & 10\sin(\theta_1)+5\sin(\theta_1+\theta_2) \\ 0 & 0 & -1 & 10-d_3 \end{vmatrix}$$

Posicion efector final

$x = 10\cos(\theta_1) + 5\sin(\theta_1+\theta_2)$
$y = 10\sin(\theta_1) + 5\sin(\theta_1+\theta_2)$
$z = 10 - d_3$

Euler lagrange

$$E_k = \frac{1}{2} m_1 V_{c1}^2 + \frac{1}{2} I_1 \dot{\theta}_1^2 + \frac{1}{2} m_2 V_{c2}^2 + \frac{1}{2} I_2 (\dot{\theta}_1+\dot{\theta}_2)^2 + \frac{1}{2} m_3 \dot{d}_3^2$$

$E_p = -m_3 g\, d_3$

Matriz Inercia

$$M(q) = \begin{vmatrix} I_1+I_2+m_1 r_1^2 + m_2(100+r_2^2) + 2\, m_2 10 r_2 \cos(\theta_2 & I_2+m_2 r_2^2 + m_2 10 r_2 \cos(\theta_2) & 0 \\ I_2+m_2 r_2^2 + m_2 a_1 r_2 + m_2 10 r_2 \cos(\theta_2) & I_2+m_2 r_2^2 & 0 \\ 0 & 0 & m_3 \end{vmatrix}$$

Matriz de coriolis

$$C = \begin{vmatrix} -h\dot{\theta}_2 \sin(\theta_2) & -h(\dot{\theta}_1+\dot{\theta}_2)\sin(\theta_2) & 0 \\ h\dot{\theta}_1 \sin(\theta_2) & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

Vector de Gravedad:

$$G(q) = \begin{vmatrix} 0 \\ 0 \\ -m_3 g \end{vmatrix}$$

$$\begin{vmatrix} q \\ \dot{q} \end{vmatrix} = \begin{vmatrix} \dot{q} \\ -M(q)^{-1}\left(C(q,\dot{q})\dot{q} + G(q)\right) \end{vmatrix} + \begin{vmatrix} 0 \\ M(q)^{-1} \end{vmatrix}$$

Si $q=0$ y la velocidad $=0$

$$A = \begin{vmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{vmatrix} \quad B = \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ & -M(q)^{-1} & \end{vmatrix} = \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0{,}036 & -0{,}107 & 0 \\ -0{,}107 & 0{,}358 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

$$C = \begin{vmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{vmatrix} \quad D = \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

Desacople

$V_1 = 1$ Para entra 1

$$\begin{vmatrix} 1 & 0 & 0 & 0 & 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0{,}036 & -0{,}107 & 0 \\ -0{,}109 & 0{,}358 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 0 \end{vmatrix}$$

$V_1 = 2$

$$\begin{vmatrix} 1 & 0 & 0 & 0 & 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0{,}036 & -0{,}107 & 0 \\ -0{,}109 & 0{,}358 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 0{,}036 & -0{,}107 & 0 \end{vmatrix}$$

$V_2 = 2$ Para entrada 2

$$\begin{vmatrix} 0 & 1 & 0 & 0 & 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0{,}036 & -0{,}107 & 0 \\ -0{,}109 & 0{,}358 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} -0{,}107 & 0{,}036 & 0 \end{vmatrix}$$

$V_3 = 2$ Para entrada 3

$$\begin{vmatrix} 0 & 0 & 1 & 0 & 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0{,}036 & -0{,}107 & 0 \\ -0{,}109 & 0{,}358 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 \end{vmatrix}$$

$$H_{2A} = \begin{vmatrix} 1/s^2 & 0 & 0 \\ 0 & 1/s^2 & 0 \\ 0 & 0 & 1/s^2 \end{vmatrix} \qquad \epsilon = \begin{vmatrix} 0{,}036 & -0{,}107 & 0 \\ -0{,}107 & 0{,}036 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$P = \begin{vmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{vmatrix} \qquad F = \epsilon^{-1} P = \begin{vmatrix} 0 & 0 & 0 & 0{,}036 & -0{,}107 & 0 \\ 0 & 0 & 0 & -0{,}107 & 0{,}036 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{vmatrix}$$

$F(s) = \frac{1}{s^2}$ $\quad e_{ss} = 0$ $\quad t_{sd} = 1$ $\quad \zeta_d = 0{,}7$ $\quad \omega_{nd} = \frac{4}{1 \cdot 0{,}7} = 5{,}71$

$T_m = \frac{1}{20} = 0{,}05$

$$\frac{0{,}00125\,z + 0{,}00125}{z^2 - 2z + 1} \Rightarrow \frac{0{,}00125\,z^{-1} + 0{,}00125\,z^{-2}}{1 - 2z^{-1} + z^{-2}}$$

FCO:

$$G = \begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} \qquad H = \begin{vmatrix} 0{,}00125 \\ 0{,}00125 \end{vmatrix}$$

$$C = \begin{vmatrix} 0 & 1 \end{vmatrix} \qquad D = 0$$

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$$\hat{S} = [\hat{H} \;\; \hat{G}\hat{H} \;\; \hat{G}^2\hat{H}] = \begin{vmatrix} 0{,}0013 & -0{,}0013 & -0{,}0037 \\ 0{,}0013 & 0{,}0037 & 0{,}0063 \\ -0{,}0016 & -0{,}005 & -0{,}0112 \end{vmatrix}$$

$(z^2 + p_1 z + p_2)(z - 0{,}1)$

$z^3 - 1{,}7z^2 + 0{,}831z - 0{,}067$

$$\phi G = \begin{vmatrix} -0,367 & -0,431 & 0 \\ 0,431 & 0,495 & 0 \\ -1,731 & -2,162 & 0 \end{vmatrix} \qquad K = \begin{vmatrix} 440,2 & 574,2 & -25,6 \end{vmatrix}$$

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$$t_{so} = \frac{1}{10} = 0,1 \qquad \omega_{ndo} = \frac{4}{0,1 \cdot 0,7} = 57,1429$$

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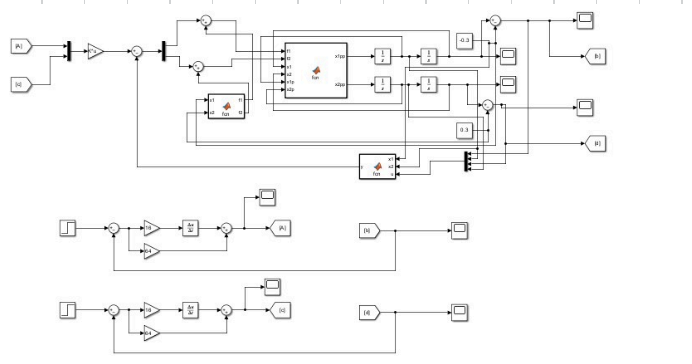
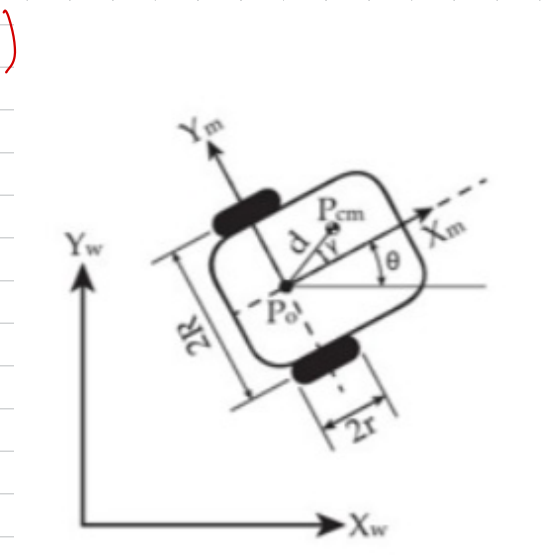
$$z^2 + 0,122\,z + 0,0183$$

$$L = \begin{vmatrix} -0,9817 \\ 2,1225 \end{vmatrix}$$



⇐ simu no lineal

4)



Modelo cinemático

Entradas ω_I, ω_D

Salidas x, y, θ

$\dot{v}(t) = \dot{x}_1(t) = x_2(t)$

$\dot{x}_2(t) = u_1 x_2(t) + b_{r1} \omega_i(t) + b_{r2} \omega_D(t)$

$\dot{\theta} = x_3(t) = x_4(t)$

$\dot{x}_4(t) = v_\theta x_4(t) + b_{\theta 1} \omega_i(t) + b_\theta \omega_D(t)$

$b(s)_1 = \dfrac{b_r}{s^2 - v_r s}$ $b(s)_2 = \dfrac{b_\theta}{s^2 + v_\theta s}$

Modelo dinámico

entradas = V_D, V_I

$V(s) = (Ls + R_m)\, I(s) + K_e\, w(s)$

$T(s) = K_t I(s) = Js\, W(s) + b w(s)$

$I(s) = \dfrac{Js + b}{K_i} \cdot w(s)$

Sustitucion en la ecuacion de voltaje

$$V(s) = (Ls + Rm) \cdot \frac{Js + b}{k_t} \cdot w(s) + k_e\, w(s)$$

$$\frac{w(s)}{V(s)} = \frac{k_t}{(Ls + Rm)(Js + b) + k_t k_e}$$

$$\frac{k_t \cdot V_I(s)}{(Ls + Rm)(Js + b) + k_t k_e} = \frac{k_t \cdot V_D(s)}{(Ls + Rm)(Js + b) + k_t k_e}$$

Desacople de estados

$$A = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & U_r & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & U_\theta \end{vmatrix} \quad B = \begin{vmatrix} 0 & 0 \\ b_{r_1} & b_{r_2} \\ 0 & 0 \\ b_{\theta_1} & b_{\theta_2} \end{vmatrix} \quad C = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{vmatrix} \quad D = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$$

$U_r = 1 \quad U_\theta = 2 \quad b_{r_1} = 3 \quad b_{r_2} = 4 \quad b_{\theta_1} = 5 \quad b_{\theta_2} = 6$

$$A = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 \end{vmatrix} \quad B = \begin{vmatrix} 0 & 0 \\ 3 & 4 \\ 0 & 0 \\ 5 & 6 \end{vmatrix} \quad C = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{vmatrix} \quad D = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$$

$V_1 = 1$ = Primera entrada

$$\begin{vmatrix} 1 & 0 & 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 \\ 3 & 4 \\ 0 & 0 \\ 5 & 6 \end{vmatrix} = \begin{vmatrix} 0 & 0 \end{vmatrix}$$

$V_1 = 2$

$$\begin{vmatrix} 1 & 0 & 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 \\ 3 & 4 \\ 0 & 0 \\ 5 & 6 \end{vmatrix} = \begin{vmatrix} 3 & 4 \end{vmatrix}$$

$V_2 = 1$ Segunda entrada

$$\begin{vmatrix} 0 & 0 & 1 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 \\ 3 & 4 \\ 0 & 0 \\ 5 & 6 \end{vmatrix} = \begin{vmatrix} 0 & 0 \end{vmatrix}$$

$V_2 = 2$

$$\begin{vmatrix} 0 & 0 & 1 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 \\ 3 & 4 \\ 0 & 0 \\ 5 & 6 \end{vmatrix} = \begin{vmatrix} 5 & 6 \end{vmatrix}$$

$$H_{LA} = \begin{vmatrix} 1/s^2 & 0 \\ 0 & 1/s^2 \end{vmatrix}$$

$$E = \begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix} \qquad D = \begin{vmatrix} C_1 A^2 \\ C_2 A^2 \end{vmatrix} = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{vmatrix}$$

$$F = E^{-1} D = \begin{vmatrix} 0 & -3 & 0 & 4 \\ 0 & 2{,}5 & 0 & -3 \end{vmatrix}$$

Desacople de estados dinámico

$$A = \begin{vmatrix} 0 & 1 & 0 & 0 \\ \frac{-R_m b + K_t \cdot K_e}{LJ} & \frac{-Lb + R_m J}{LJ} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{-R_m b + K_t \cdot K_e}{2J} & \frac{-Lb + R_m J}{LJ} \end{vmatrix} \qquad B = \begin{vmatrix} 0 & 0 \\ \frac{K_t}{LJ} & 0 \\ 0 & 0 \\ 0 & \frac{K_t}{LJ} \end{vmatrix} \qquad \begin{matrix} C = 1 \\ D = 0 \end{matrix}$$

$R_m = 5 \qquad b = K_t = K_e = J = L = 2$

$$A = \begin{vmatrix} 0 & 1 & 0 & 0 \\ -1{,}5 & 1{,}5 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1{,}5 & 1{,}5 \end{vmatrix} \qquad B = \begin{vmatrix} 0 & 0 \\ 0{,}5 & 0 \\ 0 & 0 \\ 0 & 0{,}5 \end{vmatrix}$$

$$C = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{vmatrix} \qquad D = \begin{vmatrix} 0 & 0 \end{vmatrix}$$

Con la primera entrada

$V_1 = 1 \quad = \begin{vmatrix} 0 & 0 \end{vmatrix}$

$V_2 = 2 \quad = \begin{vmatrix} 0{,}5 & 0 \end{vmatrix}$

Para entrada 2

$V_2 = 1 \quad = \begin{vmatrix} 0 & 0 \end{vmatrix}$

$V_2 = 2 \quad = \begin{vmatrix} 0 & 0{,}5 \end{vmatrix}$

$$H_{2A} = \begin{vmatrix} 1/s^2 & 0 \\ 0 & 1/s^2 \end{vmatrix}$$

$$E = \begin{vmatrix} 0{,}5 & 0 \\ 0 & 0{,}5 \end{vmatrix} \qquad P = \begin{vmatrix} C_1 A^2 \\ C_2 A^2 \end{vmatrix} = \begin{vmatrix} -1{,}5 & 1{,}5 & 0 & 0 \\ 0 & 0 & -1{,}5 & 1{,}5 \end{vmatrix}$$

$$F = \begin{vmatrix} -3 & 3 & 0 & 0 \\ 0 & 0 & -3 & 3 \end{vmatrix}$$

Retro estados

$F(s) = \dfrac{1}{s^2} \qquad \zeta d = 0{,}9 \qquad tsd = 0{,}3 \qquad T = \dfrac{0{,}3}{20} = 0{,}015$

$$\frac{0{,}0001125\,z + 0{,}0001125}{z^2 - 2z + 1}$$

$$G = \begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} = \qquad H = \begin{vmatrix} 0{,}0001125 \\ 0{,}0001125 \end{vmatrix}$$

$$C = \begin{vmatrix} 0 & 1 \end{vmatrix} \qquad D = 0$$

$$\det |H \;\; GH| = 5{,}07 \times 10^{-3}$$

$$\hat{G} = \begin{vmatrix} 0 & -1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 \\ -1 & -2 & 1 & 1 & 1 \\ -1 & -2 & 0 & 1 & 1 \\ -1 & -2 & 0 & 0 & 0 \end{vmatrix} \qquad \hat{H} = \begin{vmatrix} 0{,}000113 \\ 0{,}000113 \\ -0{,}000113 \\ -0{,}000113 \\ -0{,}000113 \end{vmatrix}$$

$t_{sd} = 0{,}3 \qquad \omega_{nd} = 14{,}81$

$\zeta_d = 0{,}9$

$P_1 = -1{,}62$

$P_2 = 0{,}67$

$$P_D = (z^2 - 1{,}62z + 0{,}67)\,(z - 0{,}1)^3$$

$$z^5 - 1{,}929z^4 + 1{,}189z^3 - 0{,}25z^2 + 0{,}021z - 0{,}00067$$

$$\hat{S} = [\hat{H} \;\; \hat{G}\hat{H} \;\; \hat{G}^2\hat{H} \;\; \hat{G}^3\hat{H} \;\; \hat{G}^4\hat{H} \;\; \hat{G}^5\hat{H}]$$

$$\hat{S} = \begin{vmatrix} 0{,}000113 & -0{,}000113 & -0{,}000338 & -0{,}000563 & -0{,}000788 \\ 0{,}000113 & 0{,}000338 & 0{,}000563 & 0{,}000788 & 0{,}001013 \\ -0{,}000113 & -0{,}000677 & -0{,}002256 & -0{,}00563 & -0{,}0118 \\ -0{,}000113 & -0{,}000562 & -0{,}001577 & -0{,}003377 & -0{,}00619 \\ -0{,}000113 & -0{,}000451 & -0{,}001013 & -0{,}001801 & -0{,}06281 \end{vmatrix}$$

$k_1 = 8039{,}22 \qquad k_{i1} = -198{,}32$

$k_2 = 14954{,}3 \qquad k_{i2} = -1301{,}5$

$k_{i3} = -4586{,}9$

Observador

$\det(zI - (G - LC))$

$LC = \begin{vmatrix} 0 & L_1 \\ 0 & L_2 \end{vmatrix}$ $\quad G - LC = \begin{vmatrix} 0 & -1-L_1 \\ 1 & 2-L_2 \end{vmatrix}$

$\det = \begin{vmatrix} z & 1+L_1 \\ -1 & z-2+L_2 \end{vmatrix}$

$z(z-2+L_2) - (-1)(1+L_1)$

$z^2 - 2z + L_2 z + 1 + L_1$

$z^2 + z(L_2-2) + L_1 + 1$

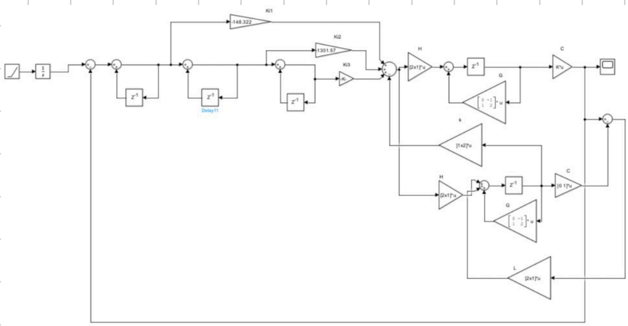
$t_{sd0} = \dfrac{0{,}3}{10} = 0{,}03$ $\quad \zeta d = 0{,}9$ $\quad \omega nd = 148{,}75$ $\quad p_1 = -0{,}1533$

$p_2 = 0{,}0183$

$z^2 - 0{,}1533\, z + 0{,}0783$

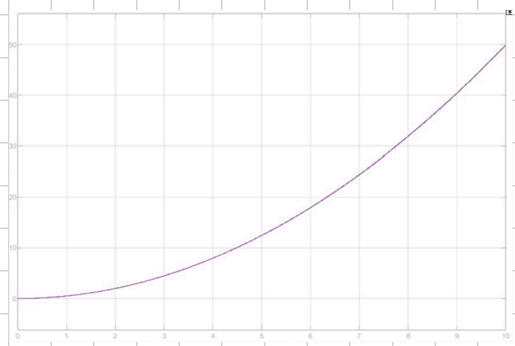
$L_2 = 1{,}8467$

$L_2 = -0{,}9817$



$\Rightarrow$ Simu con observador

Respuesta $c=$



5)

$$\begin{cases} \ddot{x}_1 + \dot{x}_1 + 5(x_1 - x_2) = f_1(t) \\ \ddot{x}_2 - 3(x_1 - x_2) + \dot{x}_2 = f_2(t) \end{cases}, \quad f_1(t), f_2(t) = \text{inputs}, x_1, x_2 = \text{outputs}$$

$$\dot{X}_1 = V_1$$
$$\dot{X}_2 = V_2$$

$$\dot{V}_1 + V_2 + 5(x_1 - x_2) = f_1(t)$$
$$\dot{V}_2 - 3(x_1 - x_2) = F_2$$

$$\dot{V}_1 = F_1 - V_1 - 5X_1 + 2X_2$$
$$\dot{V}_2 = F_2 + 3X_1 - 3X_2$$

$$\begin{bmatrix} \dot{X}_1 \\ \dot{V}_1 \\ \dot{X}_2 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -5 & -1 & 5 & 0 \\ 0 & 0 & 0 & 1 \\ 3 & 0 & -3 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ V_1 \\ X_2 \\ V_2 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{matrix} F_1 \\ F_2 \end{matrix}$$

$$y = | x_1 \quad x_2 |$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ V_1 \\ X_2 \\ V_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \end{bmatrix} F_1, F_2$$

Desacople:

$$V_1 = \begin{vmatrix} 1 & 0 & 0 & 0 \end{vmatrix} \begin{vmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 0 \end{vmatrix}$$

$$V_1 = 2 \begin{vmatrix} 1 & 0 & 0 & 0 \end{vmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ -5 & -1 & 5 & 0 \\ 0 & 0 & 0 & 1 \\ 3 & 0 & -3 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{vmatrix} 1 & 0 \end{vmatrix}$$

$$V_2 = 1 \begin{vmatrix} 0 & 0 & 1 & 0 \end{vmatrix} \begin{vmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 0 \end{vmatrix}$$

$$V_2 = 2 \begin{vmatrix} 0 & 0 & 1 & 0 \end{vmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ -5 & -1 & 5 & 0 \\ 0 & 0 & 0 & 1 \\ 3 & 0 & -3 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{vmatrix} 0 & 1 \end{vmatrix}$$

$$H_{LA} = \begin{vmatrix} 1/s^2 & 0 \\ 0 & 1/s^2 \end{vmatrix}$$

$$E = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \qquad P = \begin{vmatrix} -5 & -1 & 5 & 0 \\ 3 & 0 & -3 & 0 \end{vmatrix}$$

$$F = E^{-1} P = \begin{vmatrix} -5 & -1 & 5 & 0 \\ 3 & 0 & -3 & 0 \end{vmatrix}$$

$F(s) = \dfrac{1}{s^2}$ $\quad e_{ss} = 0$ $\quad t_{sd} = 2$ $\quad \zeta_d = 1$ $\quad \dfrac{4}{t_{sd} \cdot \zeta_d} = 2 = \omega_{nd}$

$\dfrac{2}{20} = 0{,}1$

$$\frac{0{,}005\,z + 0{,}005}{z^2 - 2z + 1} \Rightarrow \frac{0{,}005\,z^{-1} + 0{,}005\,z^{-2}}{1 - 2z^{-1} + z^{-2}}$$

FCO

$$G = \begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} \qquad H = \begin{vmatrix} 0{,}005 \\ 0{,}005 \end{vmatrix}$$

$$C = \begin{vmatrix} 0 & 1 \end{vmatrix} \qquad D = \begin{vmatrix} 0 \end{vmatrix}$$

$$\hat{G} = \begin{vmatrix} 0 & -1 & 0 \\ 1 & 2 & 0 \\ -1 & -2 & 1 \end{vmatrix} \qquad \hat{H} = \begin{vmatrix} 0{,}005 \\ 0{,}005 \\ -0{,}005 \end{vmatrix}$$

$$\hat{S} = \begin{vmatrix} 0{,}005 & -0{,}005 & -0{,}015 \\ 0{,}005 & 0{,}015 & 0{,}025 \\ -0{,}005 & -0{,}020 & -0{,}045 \end{vmatrix}$$

$\xi d = 1 \qquad Ts = 2 \qquad Tm = 0{,}1$

$$wnd = \frac{4}{1 \cdot 2} = 2$$

$$(s^2 + 2\,wnd \cdot \xi d\; s + wnd^2)$$

$$s^2 + 4s + 4$$

$s = -2$

$$P_1 = -2 \cdot e^{-\xi \cdot wnd \cdot tm} \cdot \cos(wnd \cdot tm \cdot \sqrt{1 - \xi^2}) = -1{,}6375$$
$$P_2 = e^{-2\xi \cdot wnd \cdot tm} = 0{,}6703$$

$$(z^2 + P_1 z + P_2)(z - 0{,}1)$$
$$z^3 - 1{,}74\,z^2 + 0{,}834\,z - 0{,}067$$

$$\phi G = \hat{G}^3 - 1{,}74\,\hat{G}^2 + 0{,}834 \cdot \hat{G} - 0{,}067\, I$$

$$\phi G = \begin{bmatrix} -0{,}327 & -0{,}354 & 0 \\ 0{,}354 & 0{,}381 & 0 \\ -1{,}614 & -1{,}968 & 0{,}027 \end{bmatrix}$$

$$K = \begin{vmatrix} 108{,}975 & 140{,}325 & -2{,}7 \end{vmatrix}$$

Observador

$$t_{so} = \frac{2}{10} = 0{,}2 \qquad w_{nd} = \frac{4}{\zeta \cdot t_{so}} = 20$$

$$(s^2 + 2 \cdot w_{ndo} \cdot \zeta + w_{ndo}^2) = s^2 + 40s + 400$$

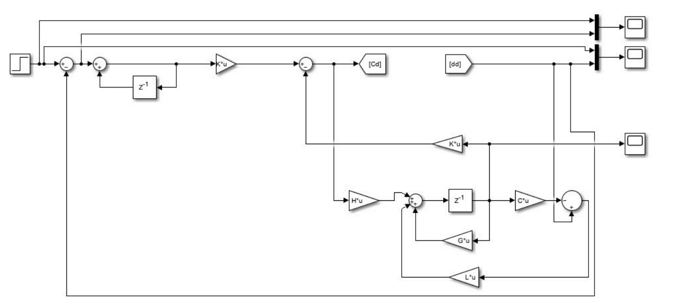
$$s = -20$$

$$P_{1o} = -0{,}2707$$

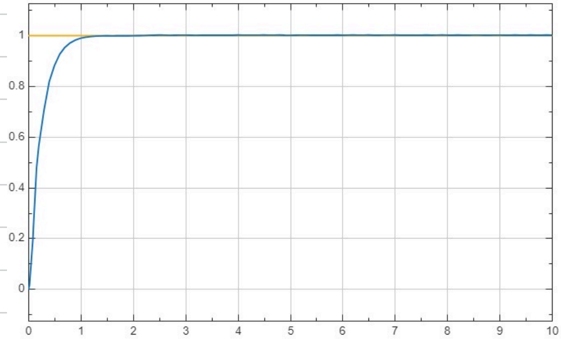
$$P_{2o} = 0{,}0183$$

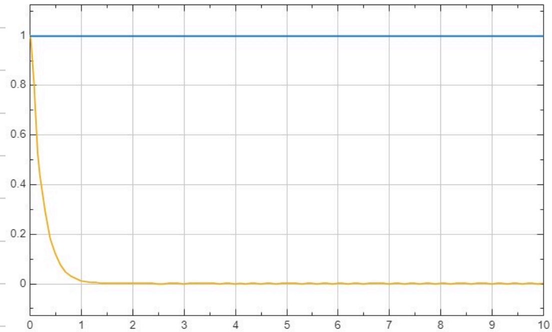
$$z^2 - 0{,}271\, z + 0{,}0183$$

$$L = \begin{vmatrix} 0{,}0183 - 1 \\ -0{,}271 + 2 \end{vmatrix} = \begin{vmatrix} -0{,}9817 \\ 1{,}7293 \end{vmatrix}$$



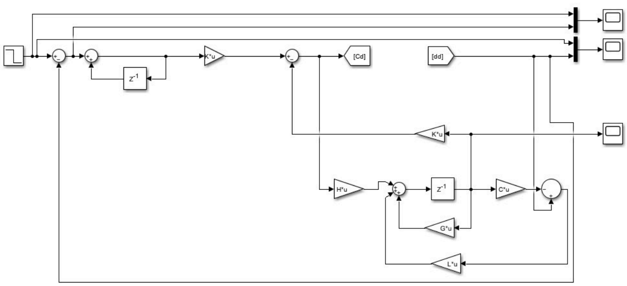
=> simu con observador y desacople



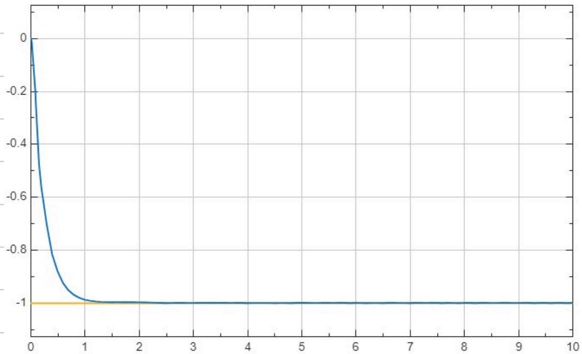


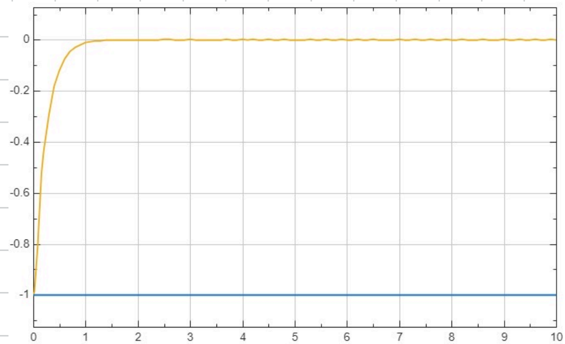
Step = 1 tsd = 1 s

error



= Simu con observadr y cambio señal





Step = 1 tsd = 1 s

error